

SOFTWARE ENGINEERING

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

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Problem Statement #36: Restaurant Table Booking & Pre-Ordering App

1. Introduction & Scenario Review

The Restaurant Table Booking & Pre-Ordering App is a restaurant hospitality application that enables diners to select specific tables using a live floor plan, place advance food orders, and calculate kitchen preparation timelines for seamless arrival dining.

ITEM	IDENTIFIED INFORMATION
Problem	Restaurant Table Booking & Pre-Ordering App
Domain	Restaurant / Hospitality
Primary Stakeholder	Diner
Other Stakeholder	Restaurant Manager
Main Functionality 1	Select specific tables
Main Functionality 2	Place advance food orders
Main Functionality 3	Calculate kitchen preparation timelines
Interface	Interactive 2D floor plan
Table Statuses	Reserved, Seated, Billed
Important Constraint	No overbooking during overlapping dining windows
Performance Requirement	Table status changes must sync across manager floor terminals in under 500 ms

2. Elicit & Draft Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow diners to pick an open table from an interactive 2D floor plan and link pre-ordered menu items to the reservation.	High	Pass: Table is reserved and kitchen pre-order ticket is queued. Fail: Overbooking of table during overlapping dining window.	Core table booking and pre-order functionality.
FR-002	Functional	The system shall allow diners to select menu items and add them to their reservation as a pre-order.	High	Pass: Selected menu items are added to the reservation with their quantities. Fail: An unavailable menu item is added to the pre-order.	Enables advance food ordering.
FR-003	Functional	The system shall allow diners to select a reservation date and dining time for an available table.	High	Pass: The reservation is created only when the selected table is available for the chosen dining period. Fail: The system creates a reservation for an overlapping dining period.	Prevents overlapping table reservations.
FR-004	Functional	The system shall calculate the estimated kitchen	High	Pass: The system displays an estimated	Helps coordinate food

		preparation time for the pre-ordered menu items.		preparation time after a pre-order is submitted. Fail: The system does not generate a preparation-time estimate.	preparation with the diner's arrival.
FR-005	Functional	The system shall allow the restaurant manager to view and update the status of restaurant tables.	Medium	Pass: The restaurant manager can update a table status and the updated status is displayed on the floor plan. Fail: An unauthorized user can update the table status.	Allows the restaurant manager to maintain accurate table status information.
NFR-001	Nonfunctional	Table status changes (Reserved, Seated, Billed) must sync across all manager floor terminals in under 500 ms.	High	Pass: Benchmarking tests confirm the target latency under simulated peak load. Fail: A table status update takes 500 ms or longer.	Ensures real-time synchronization of table status.
NFR-002	Nonfunctional	The system shall protect diner reservation and pre-order information from unauthorized access or modification.	High	Pass: Security testing confirms that unauthorized users cannot access or modify reservation and	Protects diner reservation and pre-order information.

				pre-order information. Fail: Unauthorized access or modification is possible.	
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4. Extract Actors & Use Cases:

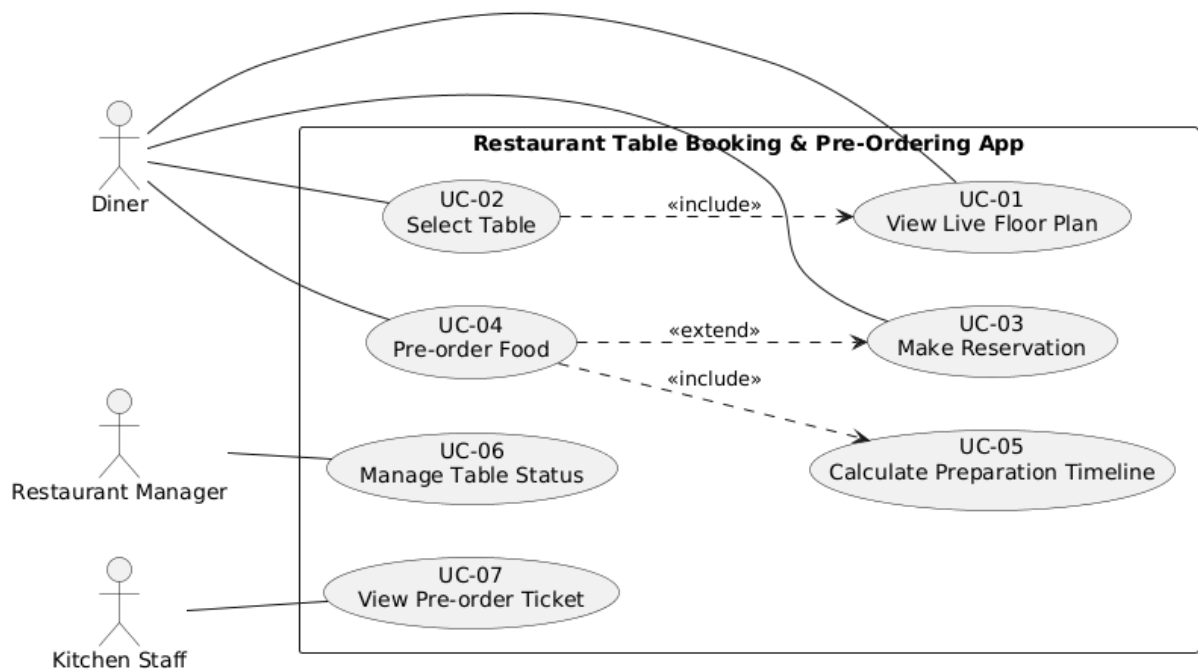
ACTORS:

ACTOR	DESCRIPTION
Diner	Uses the application to view tables, select a table, make reservations and pre-order food.
Restaurant Manager	Manages reservations and monitors/updates table statuses.
Kitchen Staff	Receives and views pre-order information for kitchen preparation.

USE CASES:

UC-01	View Live Floor Plan
UC-02	Select Table
UC-03	Make Reservation
UC-04	Pre-order Food
UC-05	Calculate Preparation Timeline
UC-06	Manage Table Status
UC-07	View Pre-order Ticket

5. UML Use-Case Diagram:



6. Use-Case Flow

Use Case: UC-03 – Make Reservation

Primary Actor: Diner

Preconditions

1. The diner has access to the Restaurant Table Booking & Pre-Ordering App.
2. The restaurant's live table information is available in the system.
3. The diner has selected or is viewing the required dining date and time.
4. At least one table is available for the selected dining period.

Postconditions

1. The selected table is reserved for the diner.
2. The reservation details are stored in the system.
3. Any pre-ordered menu items are linked to the reservation.
4. The table status is updated to Reserved.
5. A kitchen pre-order ticket is queued when food has been pre-ordered.

Main Success Scenario

1. The diner opens the Restaurant Table Booking & Pre-Ordering App.
2. The system displays the interactive 2D floor plan showing available tables.
3. The diner selects an available table.
4. The system displays the selected table and its availability.
5. The diner selects the required reservation date and dining time.
6. The system validates the availability of the selected table for the chosen dining period.
7. The diner selects menu items for advance ordering, if required.
8. The system links the selected menu items to the reservation.
9. The system calculates the estimated kitchen preparation timeline for the pre-ordered items.
10. The diner confirms the reservation.
11. The system reserves the selected table and updates its status to Reserved.
12. The system queues the kitchen pre-order ticket, if applicable.
13. The system displays the reservation confirmation.
14. The use case ends successfully.

Alternate Flow – Table Unavailable

- 4a. The system detects that the selected table is no longer available for the requested dining period.
- 4b. The system displays a message informing the diner that the selected table is unavailable.
- 4c. The system returns the diner to the live floor plan.
- 4d. The diner selects another available table.
- 4e. The system continues the reservation process from Step 5 of the Main Success Scenario.